



hyperloop
development
program

HYPERLOOP

ACCELERATING PROGRESS TOWARD
EUROPE'S GOAL OF SUSTAINABLE TRANSPORT

CONNECTING EUROPE

FASTER • CLEANER • SMARTER

HDP VISION PAPER 2026

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EXECUTIVE SUMMARY

Europe's transport system stands at a turning point. The continent has set ambitious goals for decarbonisation, connectivity, resilience, and industrial competitiveness, yet many of the instruments available today are reaching their limits. Aviation remains indispensable for long-distance mobility but is difficult to decarbonise. High-speed rail provides an efficient alternative on selected corridors, but its expansion is costly, slow, and constrained by land, infrastructure and public budgets. At the same time, passenger and freight demand continue to grow, major transport hubs are increasingly congested, and significant parts of Europe remain insufficiently connected.

Hyperloop offers Europe the possibility to add a new dimension to this system. Rather than replacing existing modes of transport, **Hyperloop** is conceived as a fifth mode that can complement rail, road, aviation, and waterways. By combining very high ground speeds with low energy consumption, electric operation, high capacity and direct point-to-point connectivity, it has the potential to close a gap that no existing mode fully addresses: fast, sustainable and scalable mobility over medium and long distances. Its ability to operate at aviation-like speeds, while retaining many of the characteristics of rail, could fundamentally change the geography of European mobility.

The opportunity extends far beyond faster journeys. A **European Hyperloop network** could relieve pressure on congested airports, rail corridors and highways, absorb part of the future growth in passenger and high-value freight transport, improve access to regions outside today's principal transport axes and materially contribute to Europe's climate objectives. Its comparatively limited land requirements and the possibility of integrating infrastructure along existing transport corridors could also make new capacity easier to deploy where conventional infrastructure expansion is increasingly difficult.

For Europe, however, **Hyperloop** is not only a transport project. It is also an industrial opportunity. European companies, universities, and research institutions have developed many of the technologies on which the concept is based, and Europe today hosts some of the world's most advanced **Hyperloop** test facilities. Moreover, the German Government's decision to establish a reference track creates the opportunity to move from subsystem development towards full system integration.

The next phase therefore requires a change in scale and organisation. After more than a decade of technological development, the central challenge is no longer simply to prove

individual components, but to integrate them into a complete transport system, demonstrate their performance at relevant speeds and operating conditions, establish the necessary safety and operational framework, and create a credible pathway towards certification and commercial deployment.

This Vision Paper proposes such a pathway.



1) WHAT IS HYPERLOOP?

Hyperloop is a new transport mode consisting of autonomous vehicles using magnetic levitation that move by magnetic propulsion in a near-vacuum tube environment. It has no mechanical friction and very small air and magnetic drag. That is why **Hyperloop** can be operated with a very low input of energy, does not cause emissions and can travel at speeds comparable to aeroplanes.

Hyperloop is therefore distinct from all existing transport modes.

More in detail, it is a public transport system capable of transporting passengers and/or high value light freight. Early versions of the system can be operated at cruising speeds of **400 - 500 km/h** by leveraging mostly existing technology. Studies such as **Hyper4Rail** indicate that the system can be operated with the most advantageous economic conditions at speeds up to **1000 km/h** and over long distances above **800 km**. To reach these higher speeds, however, significant **Test & Development (T&D)** is still needed.

Depending on the preferred alignment, the system can enter urban cores at reduced speeds to connect major multimodal hubs such as central stations. It can however also connect to public transport at the periphery of agglomerations, thus avoiding very costly construction into city centres.

It is a system in which passengers board by **Walk On Walk Off (WOWO)** and on to which freight can be loaded in standardised containers and pallet formats used for airfreight and courier traffic. It is designed to comply with rail-like acceleration and braking profiles, enabling passengers to use the vehicle without wearing seat belts.

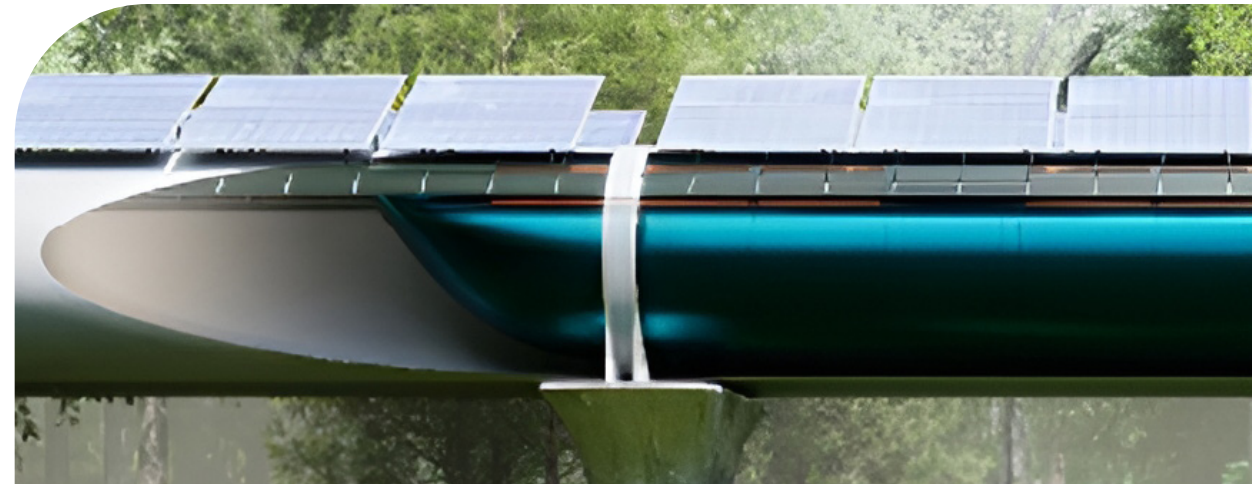
It will be failsafe in all “**reasonably foreseeable failure modes**”, being designed with multiple redundant systems. Failsafe is defined as always coming to rest at a station or pre-defined evacuation point, where passengers can disembark on foot in normal atmospheric conditions. It is designed to meet **EASA** standards for commercial aircraft, which stipulate that “catastrophic failure” per vehicle must be statistically expected to occur only once in **114.000 years of 24/7/365** operation of that vehicle.

The system is designed never to physically touch the tubular guideway whilst in motion under normative operating conditions thanks to the use of magnetic levitation. It will also be propelled, guided and controlled by electromagnetic means (linear motor). In addition, the vehicles will be capable of switching from one route to another without the need of a moving switch in the infrastructure, which would otherwise decrease the throughput capacity of the system.

The core technologies have been developed by knowledge-intensive EU companies and university R&D programmes from a concept originally proposed in Germany by **Hermann Kemper** in the 1930s and popularised as “**Hyperloop**” by **Elon Musk** in 2013. A major feasibility study into tunnelled Maglev, called “**Swissmetro**”, was also carried out in Switzerland in the 1990s.

Because switching is handled by the vehicle rather than by a moving track element, it can be performed at cruising speed, with no deceleration or recovery time needed between vehicles. Unlike rail, where the whole train must stop at every station, each **Hyperloop** vehicle is routed independently and travels directly to its destination without a mechanical switch needing to be set. This allows additional stations to be served along a corridor without adding travel time for passengers on longer trips.

Hyperloop is a fifth mode of transport, which can complement road, rail, air transport and waterways.



2) WHAT CAN HYPERLOOP OFFER? THE SCALE OF THE OPPORTUNITY.

The main value proposition of **Hyperloop** lies in its ability to provide competitive or shorter travel times compared to aviation, while simultaneously offering the capacity, comfort and sustainability of rail travel. This innovative technology fosters seamless connectivity across the continent, enabling passengers to reach **90% of EU destinations in three hours or less** and connect major centres within a country in **less than an hour** (e.g. Paris-Lyon, Berlin-Munich, Milan-Rome), while significantly reducing energy consumption compared to traditional air travel.

Its inherent multi-country and cross-border nature aligns perfectly with the EU's vision for a well-connected **TEN-T network**.

This means in particular:

1) Europe's innovators have developed and proven the core technologies required to provide **ultrafast ground transport (UGT)** networks which, when implemented, will have:

<i>The cruising speed of a plane</i>	500 to 1000 km/h depending on local circumstances
<i>The capacity of a train</i>	30.000 passengers per direction and per hour
<i>The convenience of a metro thanks to high-frequency service between major hubs</i>	Thanks to high-frequency service
<i>The carbon emissions of a bicycle</i>	When powered by renewable energies

2) Journeys between any two EU capitals at least two times faster than travelling by air, once time required getting to the airport and from the airport is taken into account.

3) Current transport modes fall short of delivering the decarbonisation targets required by law which mandate a **90% reduction of emissions by 2050**. Sustainably produced electricity, green hydrogen and sustainable aircraft fuels are not likely to be available in sufficient quantities to meet the demand that reaching these targets would generate. An additional transport mode with lower energy consumption and no emissions can help to achieve the transition.

Hyperloop provides an opportunity to develop a transport mode which has the lowest emissions and the lowest energy consumption per passenger kilometre travelled. **Powered 100% by electricity**, potentially generated, in part, by mounting solar panels on its own infrastructure and recovered from decelerating vehicles, hyperloop will be the most sustainable transport mode.

4) An increase in transport capacity to address the projected growth in passenger and freight transport demand. It is economically, environmentally, and commercially very challenging to accommodate this growth by expanding or extending railways, airports, and highways. **Hyperloop** trains can form and detach dynamically and autonomously. This system allows passengers to enjoy seamless point-to-point travel without intermediate stops in between.

5) **Hyperloop** could thus relieve congested infrastructures: short-haul flights congest airports; freight congests highways for passenger transport; high-speed rail lines cannot generally accommodate both passenger and freight traffic, due to the different speeds at which the trains move, and dedicated networks are difficult to realise.

6) An order of magnitude improvement in reliability: **Hyperloop** is almost totally protected from external interferences such as adverse weather conditions, and the sealed tube infrastructure entirely excludes interactions with '**cross traffic**' which can cause delay and failure in other modes.

7) A disruptive effect on the growth of large agglomerations and revitalisation of less accessible regions. Instead of cities sprawling out into the countryside, commuters could live in smaller cities outside of the agglomerations where accommodation is often less costly. Commuting times could be shorter over larger distances due to the point-to-point connectivity.

2) WHAT CAN HYPERLOOP OFFER? THE SCALE OF THE OPPORTUNITY. (CONT.)

8) Because **Hyperloop** requires less land, it can be built along existing infrastructures where space is scarce. Furthermore, it does not require infrastructure replacement as rapidly as in the case of high-speed rail because it has minimal wear and tear - the vehicles literally do not touch the infrastructure whilst in motion. As a result, capital expenditure (**Capex**) is expected to be lower and operating costs and maintenance (**OPEX**) will also be lower. This paves the way for a higher return on investment and makes operations economically viable.

9) Studies by the **EU-funded Hyper4Rail project**, by **Roland Berger and InnoEnergy** indicate that hyperloop is financially viable. The exceptional benefit to cost ratio enables countries to not fund construction (for which public budgets are not available), but merely to backstop private capital investment in **Design, Build, Finance, Operations & Maintenance (DBFOM)** of commercial networks. The exceptional operating economics enable projects to meet the critical **Debt Service Coverage Ratios (DSCR)** demanded by infrastructure finance providers. This will unlock up to 100% private finance for **Hyperloop** and unlock **Sovereign Wealth and Pension Funds 'patient capital' for a truly grand projet transforming the EU's "4Cs" - competitiveness, cohesion, connectivity, and carbon economics.**

10) The Network Business Case for **Hyperloop** in Europe conducted in the framework of the **Hyper4Rail** project and to be published soon concludes that "when fully deployed, the reference network could attract around **1.1 billion passengers per year, with more than 90% of passenger kilometres travelled on trips longer than 800 km.** This demonstrates that hyperloop primarily serves travel outside the effective range of high-speed rail and targets the same distance range as continental flights, where rail is not competitive".

11) In conclusion: **Hyperloop** not only proves to solve the capacity and sustainability challenges of the current transport system mix, but more than that: assuming that Europe now builds on the work of its innovators and invests the necessary industrial, institutional and public resources it has a once-in-a-generation opportunity to establish a leading new industry for the development and delivery of this new transport system. If deployed first in Europe, **Hyperloop** could also become a major European export product at the scale of Airbus. Other regions around the globe are actively working on hyperloop development, making it not **a question of whether hyperloop will exist, but rather when and by whom Hyperloop will be brought to the market.** Europe has a unique chance to lead this global movement and to position itself as a primary innovator and exporter of this technology.

Most of the benefits listed here would already apply as soon as the first lines exist and do not depend on the existence of a full network!



3) ALIGNMENT WITH EU POLICIES

Europe has set out an ambitious vision for the future of its transport system: a network that is sustainable, smart, resilient, inclusive and competitive. The **European Green Deal** commits the Union to climate neutrality by 2050, with a **90% reduction in transport emissions** as a central objective. The **Sustainable Mobility Strategy** translates this vision into concrete milestones, calling for a substantial shift from road and air to rail and other low-emission modes, the doubling of **HSR** traffic by 2030, tripling it by 2050, and the large-scale deployment of digital and automated mobility solutions. In parallel, the revised **TEN-T Regulation** aims to complete a seamless, multimodal core network by 2030 and an extended network by 2040, improving cross-border connectivity, ensuring interoperability, and enhancing resilience. These strategies define a common objective to create a transport system that not only reduces emissions, but also strengthens territorial cohesion, improves accessibility, and strengthens Europe’s role as a leader in sustainable mobility.

Hyperloop is ideally placed as a complement to better achieve these objectives: **HSR** has demonstrated that it can be competitive on medium distances of 300 – 800 km but is still

often not faster than the car from a passenger’s point of origin to destination, once the time taken to get to and from the high speed rail terminals is taken into account. It has been shown in the **Hyper4Rail Study**, that **Hyperloop** can attract very substantial numbers of passengers from all three other modes at distances from 500 km at speeds up from 750 km/h. In the same study it appears that **HSR** can have an important feeding role for **Hyperloop** lines.

In addition, transport is the weak point in Europe’s path to climate neutrality. The sector is responsible for about 29% of EU greenhouse gas emissions. Road transport makes up the largest share, while aviation continues to grow, adding CO2, noise and air pollution. On top of this, car-based mobility also means land-use for roads and parking, urban congestion and high accident rates. **HSR** shows that rail can be a clean and competitive alternative, but it has spread unevenly across Europe, with high costs and long delivery times, and cannot compete with air transport in terms of door-to-door travel times and comfort for distances over 800 km. All of this makes transport the structural bottleneck in Europe’s path to climate neutrality. And shows the need for new solutions to

decarbonise long-distance mobility.

Again, hyperloop offers a valuable contribution due to the absence of emissions, noise, low energy consumption and little use of land.

Finally, Europe has been facing a clear slowdown in productivity growth for many years – a trend that threatens future prosperity and competitiveness. As recognised in the **Draghi Report**, large-scale, transformative projects are essential to strengthen Europe’s role as a leader in sustainable mobility and to ensure it remains competitive internationally in the development of next-generation transport technologies.

Hyperloop offers a unique opportunity to contribute to industrial innovation, competitiveness and to build a new industrial sector of its own, but also to promote innovation in sectors such as steel, construction, electronics, and digitalization. It is not overstated to say that it could also become a European export story at the scale of Airbus.

Hyperloop is perfectly aligned with EU priorities on mobility, on innovation, industrial competitiveness, on the Green Deal and decarbonisation.



Requirements	Current benchmarks	Potential benchmarks with Hyperloop
Rail travel speed	By 2040, passenger railway lines on the core and extended core network must support trains traveling at speeds of 160 km/h or faster.	By 2060, Hyperloop lines on the core network must support vehicles traveling at speeds of 500 km/h or faster
European Rail Traffic Management System (ERTMS)	The single European signalling system will be deployed across the entire TEN-T network, enhancing rail safety and efficiency. National systems will be phased out.	A single European Hyperloop signalling system is developed and deployed across the entire TEN-T Hyperloop network, ensuring interoperability and digital from day one.
Airport connectivity	Major airports with over 12 million passengers annually must be connected by long-distance rail, making rail a competitive alternative to domestic feeder flights.	Major airports with over 12 million passengers annually must be connected by Hyperloop, making Hyperloop a competitive alternative to intra-European flights.

4) OTHER MOBILITY PROBLEMS THAT HYPERLOOP COULD ADDRESS

Growth of freight volumes: Freight is the fastest growing sector in the transport area. **Hyperloop** could absorb a share of the growth, especially high-priority freight which is always transported on the road.

Shift from road to rail: If **Hyperloop** absorbed passengers on long distance lines, capacity could be freed up on rail lines, currently one of the bottlenecks in the modal shift.

Congestion of airports: According to first results of the **Hyper4Rail** study, **Hyperloop** could reduce the demand for short-haul aviation considerably. While long-haul flights are more profitable and will remain the only viable solution for long-distance travel, short-haul aviation that could be replaced by **Hyperloop** is currently mainly responsible for the congestion at the major airports.

Decarbonization: Progress in supplying sustainable aeroplane fuels is slow and priority will be given to long-haul flights where there is no alternative to the aeroplane. The potential to replace short-haul aviation with a mode of transport that is at least as fast and much more energy-efficient can be a major contribution to decarbonisation. Equally, decarbonisation of road transport relies on sustainably produced electricity, competing with industry demand. If photovoltaic panels are mounted on the tubes, **Hyperloop** could produce much of its energy in a sustainable way.

Accelerating the building of infrastructure: Building additional infrastructure currently takes 15 to 20 years due to lengthy approval procedures. Where necessary, **Hyperloop** could be built along existing linear infrastructures, for example next to or above highways and, given that many such corridors have permits which enable supplementary construction within their boundaries, in less than ten years. Such ‘**corridor bundling**’ is likely to be the key practically achievable way to build new transport infrastructures in densely populated areas. It also allows significant cost savings which in turn would facilitate investment from private sources.

Scarcity of public budgets: In the EU, most Member States face severe budget restraints which means they can hardly maintain existing infrastructure and do not have the funds to build new infrastructure to address bottlenecks.

Hyperloop provides solutions to unresolved transport policy issues.



5) STATUS OF HYPERLOOP DEVELOPMENT

International Perspective

Hyperloop development is ongoing in Europe, the US, Canada, India, China, South Korea and Türkiye.

In Europe, **Hyperloop** developers cooperate through the **Hyperloop Development Program (see Annex 1)**, a foundation under Dutch law with a mandate to promote hyperloop at European level.

In addition, a student competition takes place every year in Europe with student teams from around the world, the **European Hyperloop Week**.

Europe

Currently, three major projects are ongoing that will document the state of play of **Hyperloop**:

- 1) The EU-funded **Hyper4Rail** project. A large consortium is working on a system description, a proposal for the most appropriate technology, a road map and a cost benefit analysis for **Europe's Rail Joint Undertaking (ERJU)**. Results will be published within a year.
- 2) A study documenting the state of play of **Hyperloop** commissioned by the **German Eisenbahnbundesamt**, to be published soon.
- 3) A follow-up study to **Hyper4Rail** focused on safety aspects of **Hyperloop** commissioned by **ERJU**.

Together, these studies will give a very clear overall picture.

Europe features two of the most advanced achievements in **Hyperloop** development:

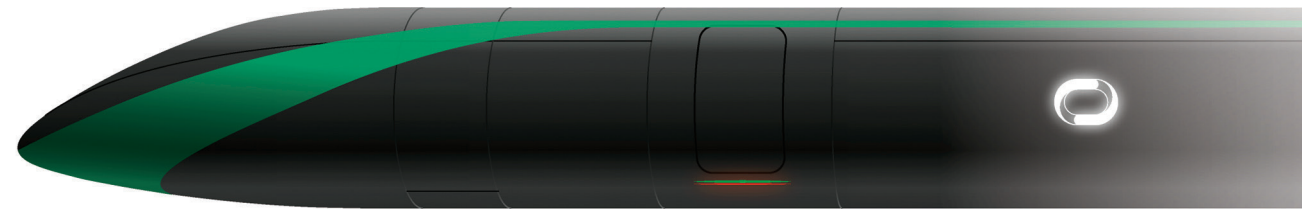
- 1) The **European Hyperloop Centre** in Veendam in the Dutch province of Groningen: **a 420-metre tube with a lane switch, the first one in the world**.

- 2) The **TUM Hyperloop** installation in Munich, Germany, with the **first full-size tube and the first ever full-size prototype vehicle**, which is certified for passenger transport under vacuum conditions.

- 3) Other test facilities include the **goTube at the University of Applied Sciences Emden/Leer** as well as **DemoTube in Zurich**, featuring a unique valve technology demonstrating the fundamentally-required capability to **isolate a tube section when circumstances require**.

The most important development however is the recent decision of the German Government to build a reference track in Germany. This project will be the first ever to fully integrate all **Hyperloop** components and allow to demonstrate proof of concept.

Europe is at the forefront of hyperloop development.



6) SYSTEM DESCRIPTION AND ONGOING DEVELOPMENT

Key vehicle parameters

Passengers will board hyperloop by walking on and walking off (**WOWO**). This will require aerospace-grade airtight, and pressure sealed doors on the vehicle and appropriate 'air dock' or 'air lock' arrangements at the station. The vehicles also include levitation and propulsion components, which allow them to move on the guideway without physical contact.

Vehicles will have a diameter comparable to a small narrow-body aircraft and will allow standard airfreight containers to be loaded. Vehicles will be at least 12 m and no more than 26 m long and designed to travel independently as well as capable of being merged into 'virtual trains'. Electronic, rather than physical coupling, enables individual vehicles to be detached from the train at high speed to stop at intermediate stations.

Key guideway parameters

The internal diameter of the guideway will be chosen appropriately to accommodate a standardise vehicle, all equipment needed outside the vehicle, and space for appropriate banking angles and aerodynamic effects. Alignments will be engineered to neutralise or mitigate the physical forces acting on passengers, to the degree required to enable **Hyperloop** to be used as a public transport system without requiring any special behaviour from, or equipment for, passengers.

It will be built from steel, concrete, or combinations of both. It will have access and egress points, as defined by the safety concept, to allow safe passenger egress, and access by authorised personnel in system failure conditions.

The **Hyper4Rail** study and other studies ongoing will provide

foundational answers to the question of whether the tubes will be built from steel or from concrete, or whether specific route sections would be more suitable for either of the two options.

Equally, the question is whether vehicles will be hanging (top levitation) or lifted from below (bottom levitation). A switch (**Zero Moving Parts Switch**) for the top levitation version is already demonstrated and tested at the **European Hyperloop Centre** and a switch for bottom levitation has been demonstrated at **TUM**.

Key safety parameters

It shall be impossible for any vehicle to collide with another or to collide with the tube or any part of the equipment contained within it.

Except under the accepted definition of a catastrophic event, as defined by **EASA** safety standards for commercial aviation (once every 114.000 years of continuous operation per vehicle) all vehicles must always execute a failsafe glide to the nearest regular station or pre-planned safe evacuation point, where passengers can disembark under normal atmospheric conditions, using door opening mechanisms which require zero external intervention, and which also prevent inadvertent opening in vacuum conditions.

All vehicles will be equipped with fully redundant emergency life support systems which cover the time taken for the failsafe glide and door opening procedure, with a safety buffer to be agreed with the regulator.

All vehicles will maintain a safe distance from each other at all times, taking into account their known locations, velocities and kinetic energy failsafe glide paths. The failsafe glide path

principle means that vehicles may follow each other at very short intervals.

Truly dynamic 'splitting and joining' whilst vehicles are in motion will ultimately be needed to fully unlock network flexibility. However, it is recognised that this will require a hyperloop specific **Operational Control System (OCS)** beyond the capacity of any current rail-derived **OCS** such as **ETCS**. This will be a key focus of the ongoing **Test & Development** programme.

Hyper4Rail will provide a complete description of the state of play of Hyperloop development and HyperSafe will develop a safety guideline to provide a framework for the certification of the technology.

7) THE ROAD MAP TO THE MARKET

After a decade of international **R&D**, various hyperloop concepts and subsystems have been developed and tested. The **Hyper4Rail** study has assessed all options and will make proposals for the best suited technology options.

It is essential that Europe organises now to seize the advantage. **HDP** and its partners will engage at senior level with relevant parties to develop a goal-focused ‘**4i**’ partnership between **Innovation, Industry, Investment, and Institutions** that is required to make **Hyperloop** deliverable, investable, fundable and buildable. This partnership will define the technical programme that is now required to Develop, Demonstrate, Derisk and Deploy **Hyperloop** both for commercial public transport in the EU and for export.

The roll-out of **Hyperloop** is expected to happen in three steps:

Step 1:

Building a 3-5 km proof of technology facility (**the German Referenzstrecke**) This addresses the **Develop, Demonstrate and Derisk** stages mentioned above.

This includes a single -or twin- tube infrastructure, low speed zero moving part switch and walk on walk off operation in at least one station demonstrator. Designed for 500 km/h in uncrewed operation and 300 km/h in crewed operation. Ideally all possible modes of construction should be included: elevated, at grade and tunnelled. It will also enable the validation of the proposed concept of operations.

Step 2:

Building a 30-50 km facility allowing certification for public service.

This infrastructure must achieve the following milestones:

- To conduct extensive (**100.000s of km**) of service simulation operations in both uncrewed and crewed modes.
- In order to achieve certification in conditions which replicate full public service, members of the public will be permitted to ride the system (**on signature of appropriate waivers**) as the

whole system approaches readiness for certification. This is represented by the public waiver milestone.

- This allows the certification milestone to be achieved.
- The final milestone is public service, and the facility could become the first commercially operating **Hyperloop** route.
- This facility must also demonstrate zero-moving-parts-switching at high speeds. In addition, it must include curved guideway to demonstrate vehicle banking which is needed for going through curves at high speed.
- This facility will trigger the **Deploy phase**.

A clear and realistic road map for reaching the deployment stage of Hyperloop exists.



8) HOW TO FINANCE HYPERLOOP DEPLOYMENT

Currently, European countries face huge challenges in maintaining their infrastructure. Public budgets are under severe strain, not least because of the need to increase defence spending. **HSR**, being the backbone of the **TEN-T network**, is very expensive to maintain and there is practically no investment in new lines, one of the big problems of implementing the **TEN-T network**. The **Rail Baltica** is a good case in point. In countries like Germany, no new infrastructure has been built in 20 years, the UK is struggling with enormous cost problems to complete the **HS2 project**, the main priority at national level. Important investments, such as the access routes to the three main alpine tunnels, the Brenner, the Gotthard and the Mont Blanc tunnel, are no longer happening. Big parts of Europe are still cut off from the **TEN-T network** and there is no perspective that this will change.

Even when only required to cover maintenance and operations, European rail companies find it difficult to be profitable, while the construction of infrastructure is at the expense of public budgets. Therefore, public policies and budgets cannot change the dramatic impact of transport on climate change and advance sufficiently to reaching the policy objectives that have been fixed.

The main question to be asked therefore to **Hyperloop** is, **can hyperloop offer a better solution?** The answer is **yes!**

How is this possible?

The answer is: **lower CAPEX and lower OPEX**, allowing private investment to build, maintain and operate **Hyperloop** lines.

CAPEX: estimations indicate that construction of a hyperloop line will not be more expensive than **HSR**. The main difference

however is that an **HSR** needs to be completely refurbished every 25 years, while a **Hyperloop** line can last 100 years, because there is no wear and tear. That means that **Hyperloop** lines will be significantly less costly over time.

In addition, **Hyperloop** lines are easy to deploy, with much less land use -not more than 2m² per linear meter when elevated, easier bridging and tunnelling (2 x 5 metres instead of 2 x 10 metres)-. They can accommodate smaller curve radii and steeper gradients. When built along or on top of existing alignments, the cost is even lower.

OPEX: the vehicles do not touch the guideway nor the infrastructure, therefore maintenance, which is one of the highest cost factors for **HSR**, is very low. Given the short journey times, there will be no staff onboard, meaning another big cost factor of **HSR** is absent. Cumulatively, this all means that hyperloop can be several times cheaper to operate than **HSR**. This in turn means that with the same ticket price, hyperloop can be much more profitable than **HSR**.

Shorter trip times, combined with the fundamental efficiency of motion in near-vacuum conditions, mean that energy efficiency will also be significantly higher, and energy costs per passenger km significantly lower.

In conclusion, lower **CAPEX** and **OPEX** will allow hyperloop to be built and operated in a much more economical way.

This means that private investors can expect to have a stable return in the long run. Government involvement could be limited to guarantees which would only be called in exceptional circumstances. **PPPs** could therefore represent win-win situations. Government funding could concentrate on maintenance of other infrastructures and completion of **TEN-T**

lines, which for many reasons may be better built as **HSR** lines.

The **Hyper4Rail** project has elaborated a **Network Business Case for Hyperloop in Europe**, which underpins the above considerations from a wider point of view: The analysis shows positive financial and economic net present values (**FNPV: €29 billion; ENPV: €657 billion**), indicating that revenues can cover long-term costs while societal benefits substantially exceed investments.

Under conservative assumptions and standard European appraisal methods, the analysis shows that such a network could be both socio-economically beneficial and financially plausible, with benefits driven mainly by travel time savings and reduced climate-related externalities.

Hyperloop as a fifth mode of transport can contribute to solving transport and climate problems without adding higher expenditure to public budgets.

9) STANDARDISATION AND REGULATION

The successful transition of **Hyperloop**, from advanced research to commercial deployment, requires a regulatory framework that is as innovative as the technology itself. This framework must strike a careful balance: it must provide the flexibility needed for technological evolution in the short term, while establishing the absolute certainty of safety and interoperability required for public service in the long term.

To achieve this, the regulatory and standardisation approach must be structured across two distinct phases:

A) The Test and Development (T&D) Phase. Pragmatically enabling standards to emerge from **T&D** operations, and to be refined and validated by operations at **European T&D facilities**.

B) The Commercial Deployment Phase. Establishing EU regulation as ‘**Gold Standard**’ for standardised, safe, and competitively interoperable **Ultra High Speed Ground Transport**.

Phase A: Test & Development

During the **Develop, Demonstrate, and Derisk phases**, it is vital that hyperloop development is granted maximum regulatory freedom. To achieve the necessary system integration, engineers and operators must have the latitude to test, iterate, and optimise without being constrained by an over-prescriptive, or premature, regulatory burden. This phase is likely to be anchored by work around system integration at the planned **Referenzstrecke facility in Germany**, which will itself build on the proofs of core technologies and subsystems already achieved at Europe’s existing **R&D facilities**.

The European Union does not begin this process from zero. Much valuable previous regulatory work carried out in Germany to establish the standards and safety case for the Transrapid maglev provides a highly relevant platform for current EU efforts to develop the next generation hyperloop regulatory, safety, and standards suite. Specifically, the **TVE Emsland maglev test facility** established a helpful precedent: it demonstrated how a dedicated test environment can successfully mature the regulatory framework for a fundamentally new, post-rail, transport system.

Looking more broadly at the ‘launch pad’ from which the upcoming phase of EU regulatory and standards development will take off, the fundamental safety of human-in-vehicle operations in

vacuum conditions at a **Hyperloop** Test and Development facility has already been empirically proven. This was demonstrated by the successful crewed test run conducted by **Josh Giegel and Sara Luchian at the Virgin Hyperloop One ‘DevLoop’ facility**. Similarly, test runs have been executed successfully at the TUM test facility based on certification by **TÜV SÜD**.

So, whilst the upcoming **Referenzstrecke** is a distinct new initiative, it will build upon these proven procedural and practical blueprints, allowing standards to organically emerge from empirical data and evolving operations.

This requires before any system enters public passenger service, it must be certified to meet or exceed the most rigorous safety benchmarks in global transport, namely **EASA standards for commercial aircraft**. To reiterate: these standards stipulate that “catastrophic failure” must be statistically expected to occur only once per vehicle in 114,000 years of 24/7/365 operation of that vehicle.

Phase B: Commercial Deployment

While the **T&D phase** requires flexibility, a clear forward vision is also required to enable the ultimate suite of regulatory and safety standards to be developed, which enables:

- All future hyperloop in the EU to follow single, unified, European standards, delivering a common infrastructure, which permits Europe-wide operation, with zero friction at national borders.
- And full competitive access to, and interoperability on, this common infrastructure by any qualified party whose equipment and operational procedures comply with that European standard.

This future objective, where regulation permits any standards-compliant vehicle to interoperate, on infrastructure built to a common single standard, with vehicles owned by other operators is the ultimate objective of hyperloop standards development. Work in the **T&D phase** must align with, and accelerate achievement of, this objective.

9) STANDARDISATION AND REGULATION (CONT.)

In view of the sheer speed and dynamic routing of **Hyperloop networks**, for which ‘**human in the loop control**’ is impossible, it is likely that Europe’s developing suite of hyperloop standards will require a completely new, highly automated control system architecture appropriate for network-scale deployment. It follows that Europe’s regulatory framework for network-scale **Hyperloop**, which will be defined and refined during the **T&D phase**, will be based upon a two-tiered control approach:

1) Tier 1 (Centralized Authority): This tier handles “**flight path allocation**” for any given start-to-stop journey profile. Because hyperloop uses multiple, smaller vehicles offering demand-responsive, single-seat service, this central layer combines baseline timetabling with real-time algorithms to ensure sufficient vehicles are at the right departure points, at the right time, across the entire network.

2) Tier 2 (Delegated Autonomy): Within the broad permissions set centrally at **Level 1**, vehicles travelling at **up to 1,000 km/h** must possess the delegated autonomy to safely execute their own trajectories. At such speeds and distances, relying on vehicle-centre-vehicle command loops for safety-critical decisions is impossible due to latency. Therefore, **Level 2** operates strictly at the vehicle level to constantly update dynamic failsafe contingency plans in **real time and absolutely guarantee zero collisions**.

It is this **Level 2** delegated autonomy that enables the regulatory cornerstone of the future safety case: the default failsafe assumption. Building directly on principles established during the maglev era, in all reasonably foreseeable failure modes (**excluding only the statistically exceptional 10^{-9} catastrophic event**), any vehicle will utilise its own kinetic energy to execute a controlled, failsafe glide to a stop at the nearest station or a pre-planned safe evacuation point. Using delegated autonomy, the regulatory standard must mandate that the entire network operates with dynamic, system-wide awareness. Should one vehicle autonomously enter a failsafe glide, every other vehicle in the system will instantaneously detect this and autonomously alter its own speed and trajectory to ensure absolute, conflict-free separation.

Finally, this forward vision extends to the commercial architecture of the network itself. To avoid the historical fragmentation of the European rail network, regulation must mandate a single, unified standard EU network architecture for **Hyperloop** infrastructure. However, this standardised infrastructure must not be an anti-competitive monopoly. The framework is designed to foster competition between operators on that shared infrastructure. The non-

negotiable “**entry ticket**” for any commercial operator to access the network will be strict, auditable compliance with unified regulatory and safety standards.

This dual-track approach is the essential catalyst for commercial realisation of **Hyperloop**.

Regulatory freedom in the **T&D phase** allows task-optimised regulatory standards to emerge today, leading to an open, ultra-safe, and interoperable European network tomorrow.

Both the **Hyper4Rail project** and the new **HyperSafe project** will provide more detailed guidance, drawing on existing standards as far as possible.



10) CHALLENGES, R&D NEEDS

The following main challenges need to be addressed on the way to commercial deployment:

A) Remaining Test & Development activities

Most subsystems exist at high **TR levels**. Open questions are:

- Testing, in particular of the switching technology, at high speeds.
- Safety protocols and installations (**Hyper4Rail** is producing these and the additional study attributed by the EU will address the remaining questions).
- Operation Control System (**ETCS** could be used up to **500 km/h**, beyond that speed a new dedicated system is to be developed).

B) System integration

This can happen once the Referenzstrecke is built in Germany.

Testing includes high speed and high-speed switching, multi-vehicle control systems, WOWO, etc. (**see attachment two: Functional specification for Germany's Referenzstrecke**).

C) Validation of Concepts of Operations.

D) Public acceptance of transport in vacuum conditions.

Hyperloop has been mainly developed by private financing. However, a stage has been reached where some public support is needed for a clearly structured **Test & Development programme** to progress the technology from its current status to be ready for commercial services. Private financing is unlikely to be available in sufficient quantity for this phase of work, which systematically overcomes the remaining challenges and eliminates the residual risk. However, once this programme has been completed, and the system achieves regulatory approval, it is highly likely that significant private capital will become available to **Design, Build, Finance, Operate & Maintain (DBFOM) commercial Hyperloop networks**. This will require the EU and member states to adopt an appropriate risk-backstopping posture in whatever public-private partnerships are developed to deliver specific projects in years to come.

Hyperloop needs a certain support from public R&D funding to reach TRL 9



11) CONCLUSION

Hyperloop is a fifth mode of transport whose core technologies built and tested by Europe's own engineers, universities and industrial partners already exist. What stands between today and commercial service is not invention, it is integration, certification, and the decision to build.

Few infrastructure projects align so many interests at once. Hyperloop offers a climate solution that does not compete for the scarce renewable electricity, green hydrogen or sustainable fuels that road and aviation decarbonisation already depend on. It offers an asset that lasts a century rather than twenty-five years, with operating economics strong enough to meet the debt service coverage ratios that unlock private and pension capital relieving, rather than adding to, public budgets already strained by defence spending and the upkeep of ageing networks. And it offers Europe the chance to write the standard for a global industry, rather than adopt one written elsewhere.

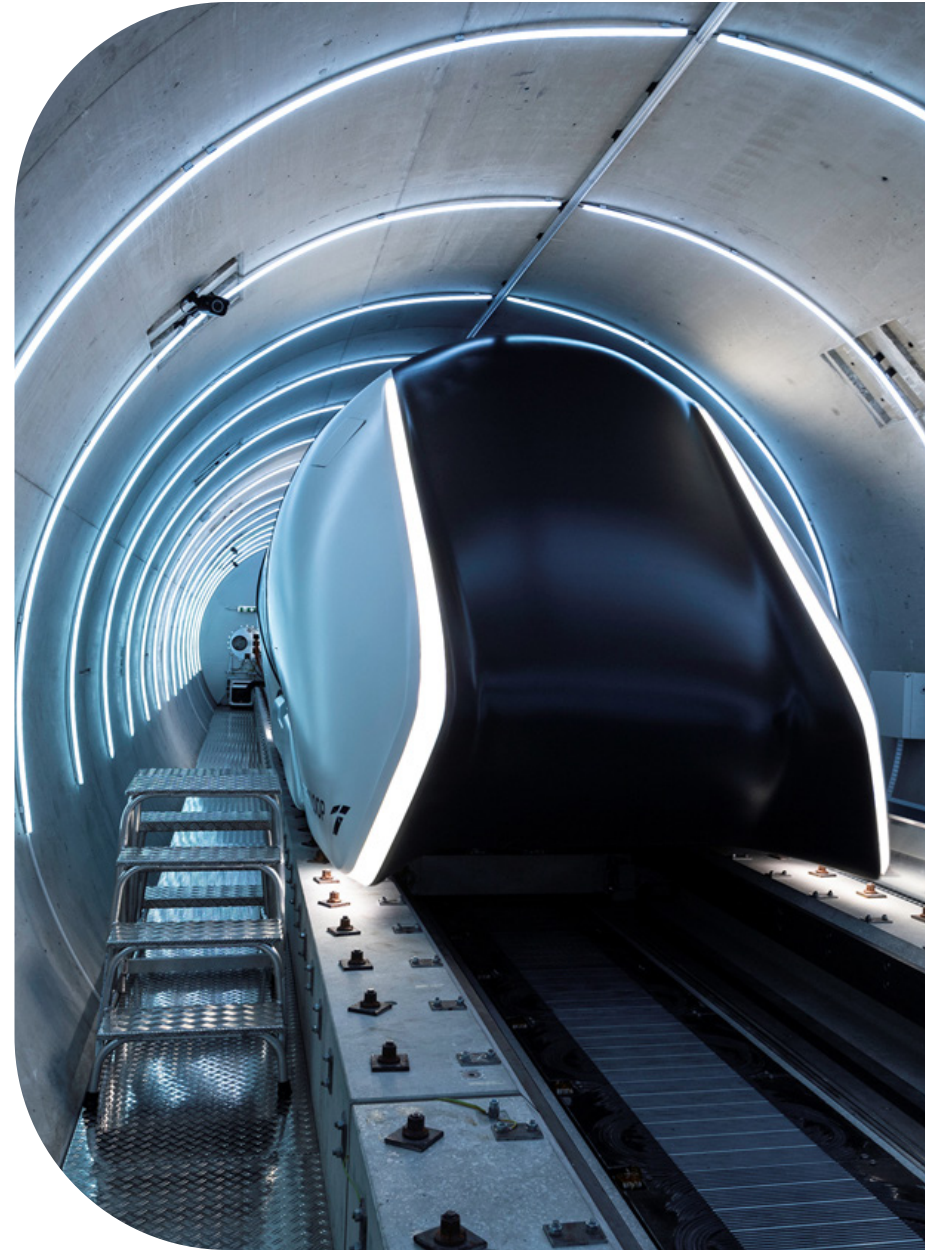
A century ago, high-speed rail redrew the map of what was reachable in a day and gave new life to regions left behind by earlier waves of infrastructure. Hyperloop can do the same for this generation, linking capitals in the time it takes to cross a city today, and giving smaller centres a genuine place in Europe's economic life rather than its periphery.

The question is no longer whether **Hyperloop** will exist. Development is underway on four continents. The question is whether it is developed, standardised and exported from Europe or imported back from wherever moves first. Europe holds the most advanced test facilities in the world and, with the German government's decision to build a reference track, the first real step towards a fully integrated demonstrator. What comes next is well defined: building on the **Referenzstrecke** and the work already done, to carry hyperloop from demonstration towards public service. It is a reasonable next step, and one Europe is well placed to take.

Annexes:

I) Mandate of HDP

II) A suggested functional specification for Germany's Referenzstrecke – HDP July 2026



ANNEX 1: HYPERLOOP DEVELOPMENT PROGRAM

The HDP is a foundation under Dutch law and has the following objectives laid down in its basic act:

- 1) Setting up innovation and development programmes and mobilising resources in cooperation with the public sector, industry and research organisations;
- 2) Support for research and development (“R&D”) programmes up to and including the level of technical readiness TRL 9 (Technology Readiness Level 9), to be further developed by commercial parties into concrete market propositions;
- 3) Facilitating public acceptance by bringing the developments taking place within the programme to the attention of a wider audience;
- 4) Creating a context in which Hyperloop can be realized;
- 5) Facilitating various test and demonstration facilities in collaboration with government and industry;
- 6) Gathering knowledge and information to support the implementation and policymaking of Hyperloop;
- 7) Stimulating and facilitating R&D activities by start-ups, scale-ups, small and medium-sized enterprises and other companies;
- 8) Stimulating and facilitating research by knowledge institutions, universities, and universities of applied sciences;
- 9) Providing space for the creation of “spin-off” and “spin-in” technologies;
- 10) Organising events for the exchange of knowledge;
- 11) Creating and fostering an open ecosystem where all partners can participate with a meaningful contribution to Hyperloop development at all levels of technical readiness (Technology Readiness Levels);
- 12) Stimulating European cooperation.

ANNEX 2: A SUGGESTED FUNCTIONAL SPECIFICATION FOR GERMANY'S REFERENZSTRECKE – JULY 2026

Recognising the importance of Germany's proposed Referenzstrecke, the members of the Hyperloop Development Programme have agreed a number of key principles, and recommend them for consideration by the German Government as it develops detailed plans for the project.

Taking into account the core technologies, sub-systems and components that have already been proven in Europe's various Hyperloop R&D facilities, HDP's consensus view is that the Referenzstrecke should ideally demonstrate the **systems integration** required to prove the key functions described below. It should deliver **proof of safe operation of the entire system** for passenger transport.

For clarity, the following assumes the Referenzstrecke operates under an appropriate level of certification for a Test & Development facility, with all human-in-system activities conducted by appropriately trained staff.

In HDP's collective view, the following major milestones would be realistically achievable, assuming the Referenzstrecke programme starts from the 2026 'state of the art' in the Hyperloop field, is constructed in the short term, and is operated for an agreed period to allow in-depth data to be gathered.

1) Walk On Walk Off (WOWO) passenger access to/from a full size Hyperloop vehicle in very low pressure operating conditions, without the passenger having to take any special measures. Essentially this means demonstrating, from a passenger perspective, that Hyperloop is as easy to use as a train.

2) Station facilities to permit WOWO operation. This means demonstrating the failsafe operation of the 'airlock' or 'airdock' subsystem required to enable passengers to embark/disembark from a Hyperloop vehicle under normal atmospheric pressure.

3) Passenger transport in a full size Hyperloop vehicle departing from and returning to a WOWO station.

4) Failsafe operation of a control system capable of realtime integration of levitation, propulsion, guidance and positional awareness. This should both demonstrate normative operations and prove a failure mode response which allows safe exit from the vehicle and the infrastructure for passengers, under normal atmospheric conditions.

5. Switching from one tube to another in a 'loop line' (Nebenstrecke) configuration. Switching is a fundamental requirement for network-scale deployment, and the 'offline loop' configuration is essential to serve intermediate stations without compromising capacity for non-stop services on the main line.

6. Simultaneous operation of at least 2 vehicles demonstrating another fundamental of public transport in network-scale application, and simultaneously proving a core requirement for the control system never to permit conflicting movement of vehicles.

ANNEX 2: A SUGGESTED FUNCTIONAL SPECIFICATION FOR GERMANY'S REFERENZSTRECKE – JULY 2026 (CONT.)

To demonstrate all the above, a set of desirable physical characteristics for the Referenzstrecke naturally emerges. A 3 km to 5 km length permits human-in-vehicle at up to 300 km/h and uncrewed operations at 500 km/h, taking acceleration and braking into account.

Demonstrating the construction techniques (especially elevated) which enable Hyperloop to follow existing transport corridors, with low land take and minimal environmental impact, may unlock a faster path for building eventual commercial routes. This also requires demonstrating the banking capability necessary to negotiate the curves found on such corridors, with an appropriate failsafe provision to avoid 'stopped on tilt' conditions.

To permit rapid progress, a control system evolved from the most advanced current ETCS in the rail sector may be usable.

However, it is noted that speeds significantly above 500 km/h may be economically advantageous for long distance Hyperloop routes. These speeds, and a possible future operating model with fully dynamic 'splitting and joining' of 'virtual trains' would require a Hyperloop-specific control system. HDP therefore recommends the Referenzstrecke adopts a **future-proofing principle** that neither it, nor the standards which will be validated by its operation, should impose any technology 'hard limits' which would close off the future upgrade path necessary to achieve operation at several hundred km/h to 1000 km/h.

If the Referenzstrecke delivers the proof points set out above, it will very significantly derisk Hyperloop as an investment proposition for the heavyweight industrial partners and infrastructure finance providers. The participation of these players is essential for network-scale commercial projects in the longer term.

The Referenzstrecke will demonstrate a pathway to low capital cost at scale, and ultra-low operation and maintenance costs, resulting from zero contact between vehicle and guideway whilst in motion. These advantages will permit private sector Design, Build, Finance, Operate & Maintain (DBFOM) for commercial routes. HDP will provide every possible assistance to ensure the Referenzstrecke succeeds in its vital mission.

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HYPERLOOP

ACCELERATING PROGRESS TOWARD
EUROPE'S GOAL OF SUSTAINABLE TRANSPORT

HDP VISION PAPER 2026

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